Large language model

A large language model (LLM) is a language model trained with self-supervised machine learning on a vast amount of text, designed for natural language processing tasks, especially language generation. The largest and most capable LLMs are generative pre-trained transformers (GPTs) that provide the core capabilities of modern chatbots. LLMs can be fine-tuned for specific tasks or guided by prompt engineering. These models acquire predictive power regarding syntax, semantics, and ontologies inherent in human language corpora, but they also inherit inaccuracies and biases present in the data they are trained on. They consist of billions to trillions of parameters and operate as general-purpose sequence models, generating, summarizing, translating, and reasoning over text. LLMs represent a significant new technology in their ability to generalize across tasks with minimal task-specific supervision, enabling capabilities like conversational agents, code generation, knowledge retrieval, and automated reasoning that previously required bespoke systems. LLMs evolved from earlier statistical and recurrent neural network approaches to language modeling. The transformer architecture, introduced in 2017, replaced recurrence with self-attention, allowing efficient parallelization, longer context handling, and scalable training on unprecedented data volumes. This innovation enabled models like GPT, BERT, and their successors, which demonstrated emergent behaviors at scale, such as few-shot learning and compositional reasoning. Reinforcement learning, particularly policy gradient algorithms, has been adapted to fine-tune LLMs for desired behaviors beyond raw next-token prediction. Reinforcement learning from human feedback (RLHF) applies these methods to optimize a policy, the LLM's output distribution, against reward signals derived from human or automated preference judgments. This has been critical for aligning model outputs with user expectations, improving factuality, reducing harmful responses, and enhancing task performance. Benchmark evaluations for LLMs have evolved from narrow linguistic assessments toward comprehensive, multi-task evaluations measuring reasoning, factual accuracy, alignment, and safety. Hill climbing, iteratively optimizing models against benchmarks, has emerged as a dominant strategy, producing rapid incremental performance gains but raising concerns of overfitting to benchmarks rather than achieving genuine generalization or robust capability improvements.

History

Before the emergence of transformer-based models in 2017, some language models were considered large relative to the computational and data constraints of their time. In the early 1990s, IBM's statistical models pioneered word alignment techniques for machine translation, laying the groundwork for corpusbased language modeling. In 2001, a smoothed n-gram model, such as those employing Kneser–Ney smoothing, trained on 300 million words, achieved state-ofthe-art perplexity on benchmark tests. During the 2000s, with the rise of widespread internet access, researchers began compiling massive text datasets from the web ("web as corpus") to train statistical language models.

Moving beyond n-gram models, researchers started in 2000 to use neural networks to learn language models. Following the breakthrough of deep neural networks in image classification around 2012, similar architectures were adapted for language tasks. This shift was marked by the development of word embeddings (eg, Word2Vec by Mikolov in 2013) and sequence-to-sequence (seq2seq) models using LSTM. In 2016, Google transitioned its translation service to neural machine translation (NMT), replacing statistical phrase-based models with deep recurrent neural networks. These early NMT systems used LSTM-based encoderdecoder architectures, as they preceded the invention of transformers.

At the 2017 NeurIPS conference, Google researchers introduced the transformer architecture in their landmark paper "Attention Is All You Need". This paper's goal was to improve upon 2014 seq2seq technology, and was based mainly on the attention mechanism developed by Bahdanau et al. in 2014. The following year in 2018, BERT was introduced and quickly became "ubiquitous". Though the original transformer has both encoder and decoder blocks, BERT is an encoder-only model. Academic and research usage of BERT began to decline in 2023, following rapid improvements in the abilities of decoder-only models (such as GPT) to solve tasks via prompting.

Although decoder-only GPT-1 was introduced in 2018, it was GPT-2 in 2019 that caught widespread attention because OpenAI claimed to have initially deemed it too powerful to release publicly, out of fear of malicious use. GPT-3 in 2020 went a step further and as of 2025 is available only via API with no offering of downloading the model to execute locally. But it was the 2022 consumer-facing chatbot ChatGPT that received extensive media coverage and public attention. The 2023 GPT-4 was praised for its increased accuracy and as a "holy grail" for its multimodal capabilities. OpenAI did not reveal the high-level architecture and the number of parameters of GPT-4. The release of ChatGPT led to an uptick in LLM usage across several research subfields of computer science, including robotics, software engineering, and societal impact work. In 2024 OpenAI released the reasoning model OpenAI o1, which generates long chains of thought before returning a final answer. Many LLMs with parameter counts comparable to those of OpenAI's GPT series have been developed. Since 2022, open-weight models have been gaining popularity, especially at first with BLOOM and LLaMA, though both have restrictions on usage and deployment. Mistral AI's models Mistral 7B and Mixtral 8x7b have a more permissive Apache License. In January 2025, DeepSeek released DeepSeek R1, a 671- billion-parameter open-weight model that performs comparably to OpenAI o1 but at a much lower price per token for users. Since 2023, many LLMs have been trained to be multimodal, having the ability to also process or generate other types of data, such as images, audio, or 3D meshes. These LLMs are also called large multimodal models (LMMs), or multimodal large language models (MLLMs).

As of 2024, the largest and most capable models are all based on the transformer architecture. Some recent implementations are based on other architectures, such as recurrent neural network variants and Mamba (a state space model). Open-weight LLMs have increasingly shaped the field since 2023, contributing to broader participation in AI development and greater transparency in model evaluation. Vake et al. (2025) demonstrated that community-driven contributions to open-weight models measurably improve their efficiency and performance, with user participation growing rapidly on collaborative platforms such as Hugging Face. Paris et al. (2025) further argued that openness in AI should extend beyond releasing model code or weights to encompass inclusiveness, accountability, and ethical responsibility in AI research and deployment. Collectively, these studies highlight that open-weight LLMs can accelerate innovation and enhance scientific reproducibility, while fostering a more transparent and participatory AI ecosystem

Dataset preprocessing

Tokenization

As machine learning algorithms process numbers rather than text, the text must be converted to numbers. In the first step, a vocabulary is decided upon, then integer indices are arbitrarily but uniquely assigned to each vocabulary entry, and finally, an embedding is associated to the integer index. Algorithms include byte-pair encoding (BPE) and WordPiece. There are also special tokens serving as

control characters, such as [MASK] for masked-out token (as used in BERT), and [UNK] ("unknown") for characters not Dataset preprocessing Tokenization appearing in the vocabulary. Also, some special symbols are used to denote special text formatting. For example, "Ġ" denotes a preceding whitespace in RoBERTa and GPT and "##" denotes continuation of a preceding word in BERT. For example, the BPE tokenizer used by the legacy version of GPT-3 would split tokenizer: texts -> series of numerical "tokens" as tokenizer: texts ->series of numerical "tokens" Tokenization also compresses the datasets. Because LLMs generally require input to be an array that is not jagged, the shorter texts must be "padded" until they match the length of the longest one. The average number of words per token depends on the language.

Byte-pair encoding

As an example, consider a tokenizer based on byte-pair encoding. In the first step, all unique characters (including blanks and punctuation marks) are treated as an initial set of n-grams (i.e. initial set of unigrams). Successively the most frequent pair of adjacent characters is merged into a bi-gram and all instances of the pair are replaced by it. All occurrences of adjacent pairs of (previously merged) n-grams that most frequently occur together are then again merged into even lengthier n-gram, until a vocabulary of prescribed size is obtained. After a tokenizer is trained, any text can be tokenized by it, as long as it does not contain characters not appearing in the initial-set of uni-grams.

Problems

A token vocabulary based on the frequencies extracted from mainly English corpora uses as few tokens as possible for an average English word. However, an average word in another language encoded by such an English-optimized tokenizer is split into a suboptimal amount of tokens. GPT-2 tokenizer can use up to 15 times more tokens per word for some languages, for example for the Shan language from Myanmar. Even more widespread languages such as Portuguese and German have "a premium of 50%" compared to English.

Dataset cleaning

In the context of training LLMs, datasets are typically cleaned by removing low-quality, duplicated, or toxic data. Cleaned datasets can increase training efficiency and lead to improved downstream performance. A trained LLM can be used to clean datasets for training a further LLM. With the increasing proportion of LLM-generated content on the web, data cleaning in the future may include filtering out such content. LLM-generated content can pose a problem if the content is similar to human text (making filtering difficult) but of lower quality (degrading performance of models trained on it).

Synthetic data

Training of largest language models might need more linguistic data than naturally available, or that the naturally occurring data is of insufficient quality. In these cases, synthetic data might be used. Microsoft's Phi series of LLMs is trained on textbook-like data generated by another LLM.